A COUNTEREXAMPLE TO KOUROVKA PROBLEM 20.28

ABSTRACT. We give a negative answer to Kourovka Problem 20.28. The group

$$G = \{A \in \mathrm{GL}_2(5) : \det A \in (\mathbb{F}_5^\times)^2\}$$

has the same set of conjugacy-class sizes as $\mathrm{SL}_2(5)$, namely $\{1, 12, 20, 30\}$, but is not $\mathrm{SL}_2(5) \times A$ for any abelian group A. Thus the problem fails already for $L = A_5$, and a published characterization of $\mathrm{SL}_2(5)$ is false as stated. More generally, the same central enlargement gives a counterexample for every nonabelian finite simple group with nontrivial Schur multiplier.

For a finite group X, write

$$\mathrm{cs}(X) = \{|x^X| : x \in X\}.$$

Kourovka Problem 20.28 asks whether, for a nonabelian finite simple group L with universal perfect central extension $H(L)$, the equality $\mathrm{cs}(G) = \mathrm{cs}(H(L))$ forces $G \cong H(L) \times A$ for some abelian group A [3, Problem 20.28]. Gorshkov asserted this for $L = A_5$, where $H(L) \cong \mathrm{SL}_2(5)$ [1, Theorem 1].

Theorem 1. *Let*

$$G = \{A \in \mathrm{GL}_2(5) : \det A \in \{1, 4\}\}.$$

Then

$$\mathrm{cs}(G) = \mathrm{cs}(\mathrm{SL}_2(5)) = \{1, 12, 20, 30\},$$

but $G \not\cong \mathrm{SL}_2(5) \times A$ for any abelian group A. Consequently, Kourovka Problem 20.28 has a negative answer, already for $L = A_5$, and [1, Theorem 1] *is false as stated.*

The construction is a central enlargement within an isoclinism family. We record it in a form that applies to every nontrivial Schur multiplier.

Proposition 2. *Let H be a finite perfect group with $Z(H) \neq 1$. Choose a prime $p \mid |Z(H)|$ and an element $z \in Z(H)$ whose order p^a is maximal among the orders of the p-elements of $Z(H)$. Let $C = \langle c \rangle \cong C_{p^{a+1}}$, and put*

$$E = \frac{H \times C}{\langle (z, c^{-p}) \rangle}.$$

Then

$$\mathrm{cs}(E) = \mathrm{cs}(H), \qquad |E| = p|H|, \qquad E' = H,$$

and $E \not\cong H \times A$ for any abelian group A.

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Proof. The natural map $H \to E$ is injective, so we identify H with its image. Let t be the image of $(1, c)$. Then

$$t^p = z, \qquad |t| = p^{a+1}, \qquad E = H\langle t\rangle, \qquad H \cap \langle t\rangle = \langle z\rangle.$$

In particular, $|E| = p|H|$. Since t is central, for $h \in H$ and $j \in \mathbb{Z}$,

$$C_E(ht^j) = C_H(h)\langle t\rangle.$$

The intersection of the two factors on the right is $\langle z\rangle$, so

$$|C_E(ht^j)| = p|C_H(h)| \quad \text{and hence} \quad |(ht^j)^E| = |h^H|.$$

Thus $\mathrm{cs}(E) = \mathrm{cs}(H)$.

An element ht^j is central in E exactly when $h \in Z(H)$, and therefore

$$Z(E) = Z(H)\langle t\rangle.$$

Also E/H is cyclic, so $E' \leq H$; since H is perfect, $H = H' \leq E'$, whence $E' = H$.

If $E \cong H \times A$ with A abelian, then the order equality forces $A \cong C_p$. By the choice of z, the largest order of a p-element of $Z(H) \times C_p$ is p^a, whereas $t \in Z(E)$ has order p^{a+1}. The centers are therefore not isomorphic, a contradiction. □

Remark 3. The groups H and E in Proposition 2 are isoclinic in the sense of Hall [2]: one has $E/Z(E) \cong H/Z(H)$, $E' = H$, and the commutator maps agree. The calculation above makes the equality of class-size sets explicit. It does not preserve multiplicities: every conjugacy class of H gives p conjugacy classes of E of the same size.

Proof of Theorem 1. Set $H = \mathrm{SL}_2(5)$, $z = -I$, and $s = 2I$. Then

$$s^2 = z, \qquad |s| = 4, \qquad \det s = 4.$$

Since $(\mathbb{F}_5^\times)^2 = \{1, 4\}$, every element of G with determinant 4 lies in Hs. Hence

$$G = H\langle s\rangle, \qquad H \cap \langle s\rangle = \langle z\rangle,$$

and therefore

$$G \cong \frac{H \times C_4}{\langle (z, c^2)\rangle}.$$

Proposition 2, with $p = 2$ and $a = 1$, gives $\mathrm{cs}(G) = \mathrm{cs}(H)$ and excludes an isomorphism $G \cong H \times A$.

It remains only to record $\mathrm{cs}(H)$. We have $|H| = 120$. For a noncentral $h \in H$, the characteristic polynomial is of exactly one of the following types. If it has a repeated root $\varepsilon = \pm 1$, then $h = \varepsilon I + N$ with $N^2 = 0 \neq N$; its centralizer algebra is $\{aI + bN : a, b \in \mathbb{F}_5\}$, and the determinant-one units number $2 \cdot 5 = 10$. If the polynomial splits over $\mathbb{F}_5$ with distinct roots, then $C_H(h)$ is a split torus of order 4. If it is irreducible over $\mathbb{F}_5$, then $\mathbb{F}_5[h] \cong \mathbb{F}_{25}$, and $C_H(h)$ is the kernel of the norm $\mathbb{F}_{25}^\times \to \mathbb{F}_5^\times$, so $|C_H(h)| = 6$. All three cases occur, for example for

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \qquad \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix},$$

respectively. Thus the noncentral class sizes are $120/10 = 12$, $120/6 = 20$, and $120/4 = 30$, proving

$$\mathrm{cs}(H) = \{1, 12, 20, 30\}.$$

For later comparison, Proposition 2 also gives

$$G' = H, \qquad Z(G) = \langle s \rangle \cong C_4.$$

Every nontrivial subgroup of $Z(G)$ contains $s^2 = z \in G'$. Hence G has no nontrivial abelian direct factor. □

Corollary 4. *Let L be a nonabelian finite simple group with nontrivial Schur multiplier $M(L)$, and let $H(L)$ be its universal perfect central extension. Then there is a finite group G_L such that*

$$\mathrm{cs}(G_L) = \mathrm{cs}(H(L)), \qquad G_L \ncong H(L) \times A$$

for any abelian group A.

Proof. The group $H(L)$ is perfect and $Z(H(L)) = M(L) \neq 1$. Apply Proposition 2. □

The gap in the published proof

At the end of the proof of [1, Theorem 1], a normal subgroup $R \le Z(T)$ with $T/R \cong A_5$ is obtained. The proof then uses the absence of abelian direct factors to identify R as a Schur-multiplier kernel. A covering kernel must, however, satisfy the stem condition

$$R \le T'$$

in addition to centrality. The example above realizes the missing case: with $T = G$ and $R = Z(G)$,

$$T/R \cong A_5, \qquad R \cong C_4, \qquad T' = \mathrm{SL}_2(5), \qquad R \not\le T',$$

while T has no nontrivial abelian direct factor. Thus the final inference does not apply.

References

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